

Task 1: Machine Learning Algorithms

1. Logistic Regression

What it is: A classification algorithm the name "regression" used to predict categorical outcomes, typically binary (e.g., Pass/Fail, Yes/No).

How it works: It calculates the probability that a given input belongs to a particular class, using a mathematical function (the sigmoid function) that maps any input to a value between 0 and 1. If the probability is above a threshold (usually 0.5), it predicts one class; otherwise, the other.

Where it's used: Email spam detection, medical diagnosis (disease/no disease), credit approval.

Real-world application: Predicting whether a bank loan applicant will default or not, based on features like income, credit score, and loan amount.

Advantages: Simple, fast to train, easy to interpret, works well for linearly separable data.

Limitations: Struggles with complex, non-linear relationships between features and outcomes.

2. Decision Tree Classifier

What it is: A classification algorithm that splits data into branches based on feature values, forming a tree-like structure of decisions, ending in a predicted class.

How it works: At each node, the tree picks the feature and threshold that best separates the data into distinct classes (based on measures like Gini impurity or entropy). This

continues until the data is well-separated or a stopping condition is met.

Where it's used: Customer segmentation, fraud detection, medical diagnosis.

Real-world application: Used in this internship's own mini projects — predicting student Pass/Fail based on Marks and Attendance.

Advantages: Easy to understand and visualize, handles both numerical and categorical data, doesn't require feature scaling.

Limitations: Prone to overfitting (memorizing training data too closely), can be unstable — small data changes can produce a very different tree.

3. K-Nearest Neighbors (KNN)

What it is: A classification algorithm that predicts a data point's class based on the classes of its "k" closest neighbors in the dataset.

How it works: When predicting a new point, KNN calculates the distance (usually Euclidean) between that point and all training points, finds the "k" nearest ones, and assigns the class that's most common among those neighbors.

Where it's used: Recommendation systems, image recognition, pattern recognition.

Real-world application: Recommending similar products on e-commerce sites based on what similar customers purchased.

Advantages: Simple to understand and implement, no training phase needed (it's a "lazy learner").

Limitations: Slow for large datasets (must compare against all points), sensitive to irrelevant features and the choice of "k."

4. Linear Regression

What it is: A regression algorithm used to predict a continuous numerical value based on one or more input features.

How it works: It finds the best-fitting straight line through the data points by minimizing the difference between actual and predicted values (typically using least squares), representing the relationship as: $y = mx + b$ (for one feature).

Where it's used: Price prediction, sales forecasting, risk assessment.

Real-world application: Predicting house prices based on features like square footage, number of rooms, and location.

Advantages: Simple, fast, easy to interpret the relationship between features and outcome.

Limitations: Assumes a linear relationship between features and outcome, which real-world data doesn't always have; sensitive to outliers.